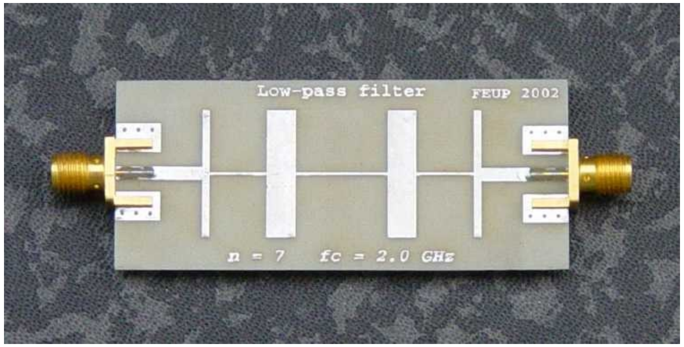
# ECE 3317

Dr. Jiefu Chen

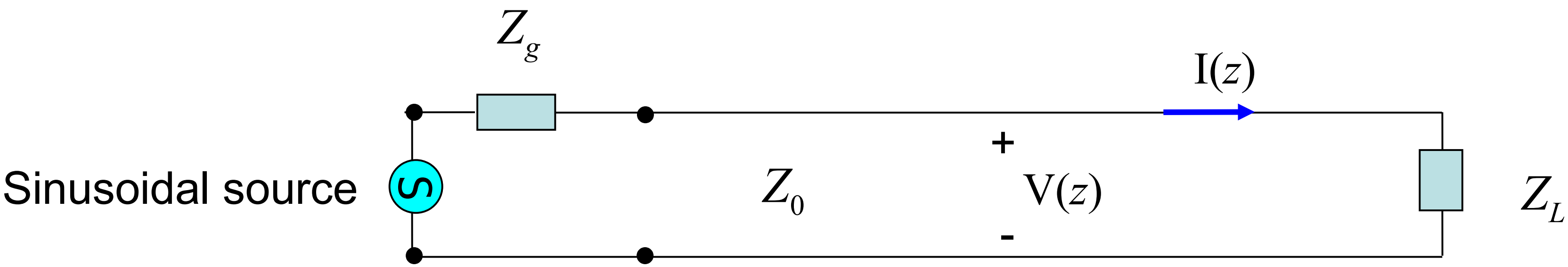
# Notes 10
# Transmission Lines
# (Reflection and Impedance)



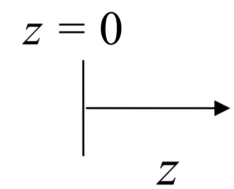

Adapted from notes by Prof. Stuart A. Long

# Reflection

Consider a transmission line that is terminated with a load:




$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$




Voltage and current on the line:

$$\mathrm{V}(z) = A e^{-\gamma z} + B e^{+\gamma z}$$

$\mathrm{V}^{+}(z)$ $\qquad$ $\mathrm{V}^{-}(z)$

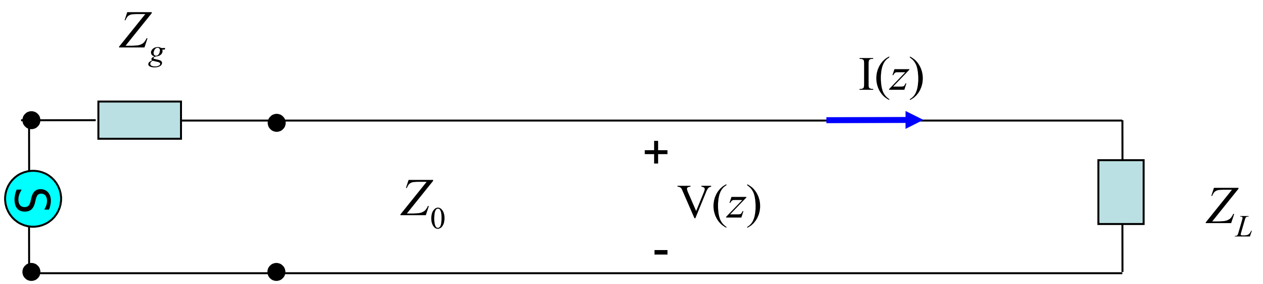
$$\mathrm{I}(z) = \left(\frac{A}{Z_0}\right) e^{-\gamma z} - \left(\frac{B}{Z_0}\right) e^{+\gamma z}$$

$\mathrm{I}^{+}(z)$ $\qquad$ $\mathrm{I}^{-}(z)$

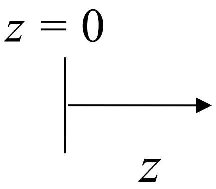
$A$ = amplitude of net forward-traveling wave
$B$ = amplitude of net backward-traveling wave

(sinusoidal steady-state)

# Reflection (cont.)




$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$




At the load ($z = 0$): $\quad \mathrm{V}(0) = Z_L\, \mathrm{I}(0)$

Hence we have $\quad A + B = \dfrac{Z_L}{Z_0}(A - B)$

or $\quad B\left(1 + \dfrac{Z_L}{Z_0}\right) = A\left(\dfrac{Z_L}{Z_0} - 1\right)$

# Reflection Coefficient

We define the load reflection coefficient: $$\Gamma_L \equiv \frac{V^-(0)}{V^+(0)}$$

$$V(z) = Ae^{-\gamma z} + Be^{+\gamma z}$$

$$V^+(z) = Ae^{-\gamma z}$$

$$V^-(z) = Be^{+\gamma z}$$

Hence $$\Gamma_L = \frac{B}{A}$$

We then have $$\Gamma_L = \frac{\left(\frac{Z_L}{Z_0} - 1\right)}{\left(1 + \frac{Z_L}{Z_0}\right)}$$

or $$\boxed{\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}}$$

# Voltage and Current

We can then write

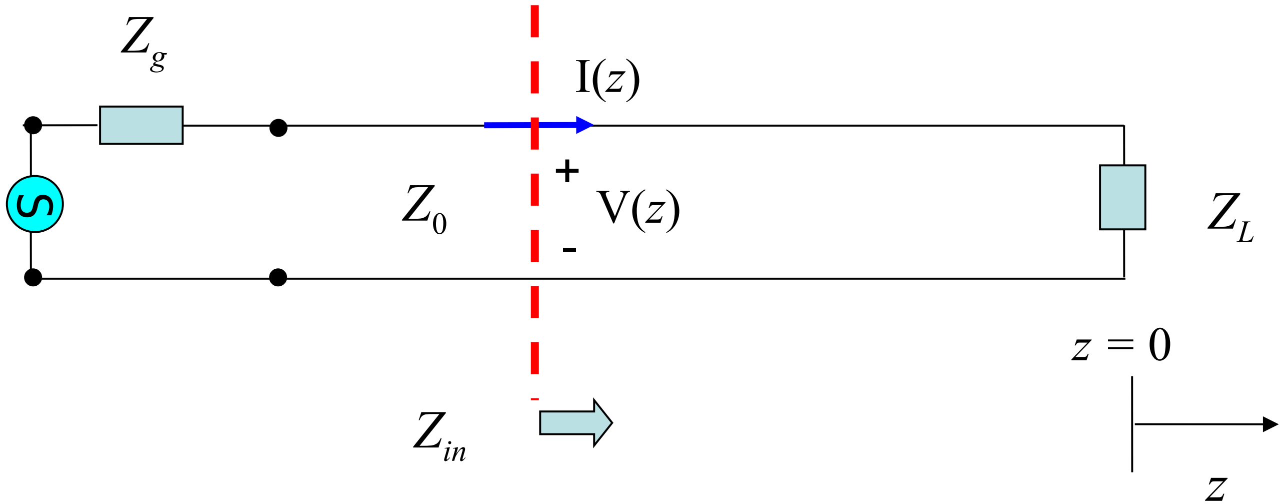
$$V(z) = A\left(e^{-\gamma z} + \Gamma_L e^{+\gamma z}\right)$$

$$I(z) = \frac{1}{Z_0} A\left(e^{-\gamma z} - \Gamma_L e^{+\gamma z}\right)$$

Note: The generator (source) will determine the unknown (complex) constant $A$.

# Impedance

We define the input impedance at any point on the line




$$Z_{in}(z) = \frac{\mathrm{V}(z)}{\mathrm{I}(z)}$$

# Impedance (cont.)

We then have

$$Z_{in}(z) = \frac{A\left(e^{-\gamma z} + \Gamma_L e^{+\gamma z}\right)}{\frac{1}{Z_0} A\left(e^{-\gamma z} - \Gamma_L e^{+\gamma z}\right)}$$

Canceling the constant $A$, we have

$$Z_{in}(z) = Z_0 \left( \frac{e^{-\gamma z} + \Gamma_L e^{+\gamma z}}{e^{-\gamma z} - \Gamma_L e^{+\gamma z}} \right)$$

or

$$Z_{in}(z) = Z_0 \left( \frac{1 + \Gamma_L e^{+2\gamma z}}{1 - \Gamma_L e^{+2\gamma z}} \right)$$

# Impedance (cont.)

$$Z_{in}(z)=Z_0\left(\frac{e^{-\gamma z}+\Gamma_L e^{+\gamma z}}{e^{-\gamma z}-\Gamma_L e^{+\gamma z}}\right)=Z_0\left(\frac{e^{-\gamma z}+\left(\frac{Z_L-Z_0}{Z_L+Z_0}\right)e^{+\gamma z}}{e^{-\gamma z}-\left(\frac{Z_L-Z_0}{Z_L+Z_0}\right)e^{+\gamma z}}\right)$$

$$=Z_0\left(\frac{(Z_L+Z_0)e^{-\gamma z}+(Z_L-Z_0)e^{+\gamma z}}{(Z_L+Z_0)e^{-\gamma z}-(Z_L-Z_0)e^{+\gamma z}}\right)$$

$$=Z_0\left(\frac{Z_L\left(e^{+\gamma z}+e^{-\gamma z}\right)-Z_0\left(e^{+\gamma z}-e^{-\gamma z}\right)}{-Z_L\left(e^{+\gamma z}-e^{-\gamma z}\right)+Z_0\left(e^{+\gamma z}+e^{-\gamma z}\right)}\right)$$

$$=Z_0\left(\frac{Z_L(2\cosh(\gamma z))-Z_0(2\sinh(\gamma z))}{-Z_L(2\sinh(\gamma z))+Z_0(2\cosh(\gamma z))}\right)$$

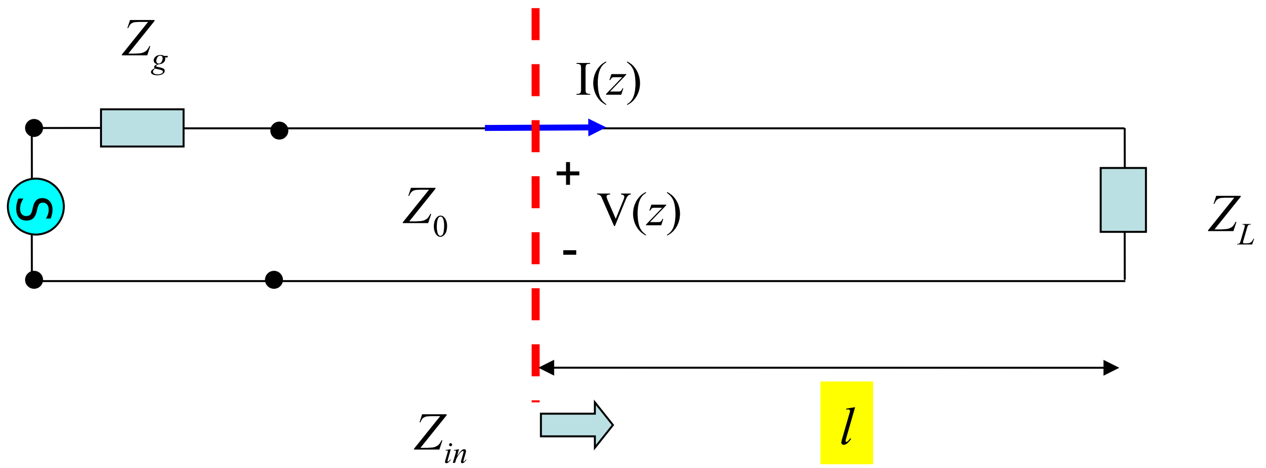
$$=Z_0\left(\frac{Z_L-Z_0\tanh(\gamma z)}{Z_0-Z_L\tanh(\gamma z)}\right)$$

Now let $z=-l$ $\quad \tanh(\gamma z)=\tanh(-\gamma l)=-\tanh(\gamma l)$

# Impedance (cont.)

We then have

$$\boxed{Z_{in} = Z_0 \left( \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right)}$$



# Impedance (cont.)

## Limiting cases

1) no transmission line:

$$l = 0 \quad \rightarrow \quad \tanh(\gamma l) = 0$$

$$Z_{in} = Z_0 \left( \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right) = Z_L$$

2) infinitely long transmission line:

$$l \rightarrow \infty \quad \rightarrow \quad \tanh(\gamma l) \rightarrow 1$$

$$Z_{in} = Z_0 \left( \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} \right) = Z_0$$

# Impedance (cont.)

Lossless Case

$$\gamma = \alpha + j\beta = \sqrt{(\cancel{R} + j\omega L)(\cancel{G} + j\omega C)} = j\omega\sqrt{LC}$$

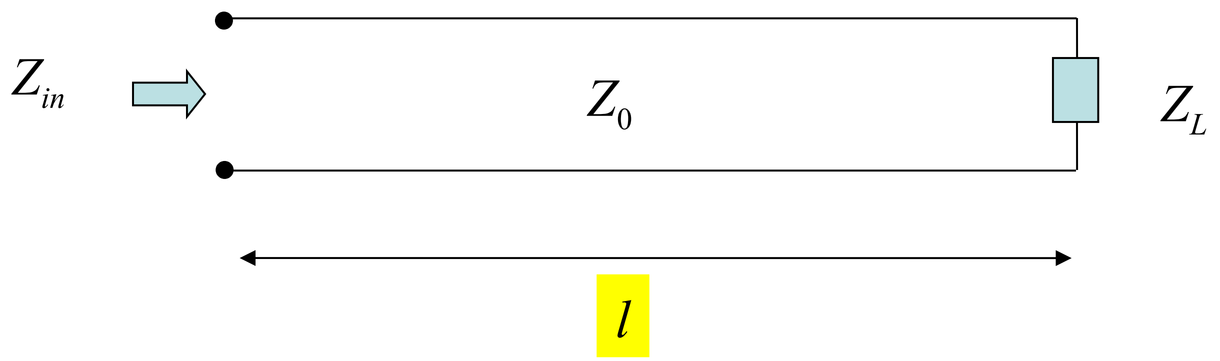
Use $\tanh(jx) = j\tan(x)$

We then have $$Z_{in} = Z_0 \left( \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right)$$

where $\beta = \omega\sqrt{LC} = \omega\sqrt{\mu\varepsilon} = \dfrac{2\pi}{\lambda}$ $\qquad \lambda = \dfrac{\lambda_0}{\sqrt{\varepsilon_r}}$

# Impedance (cont.)

Summary of final formula for a lossless line




$$Z_{in} = Z_0 \left( \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)} \right)$$

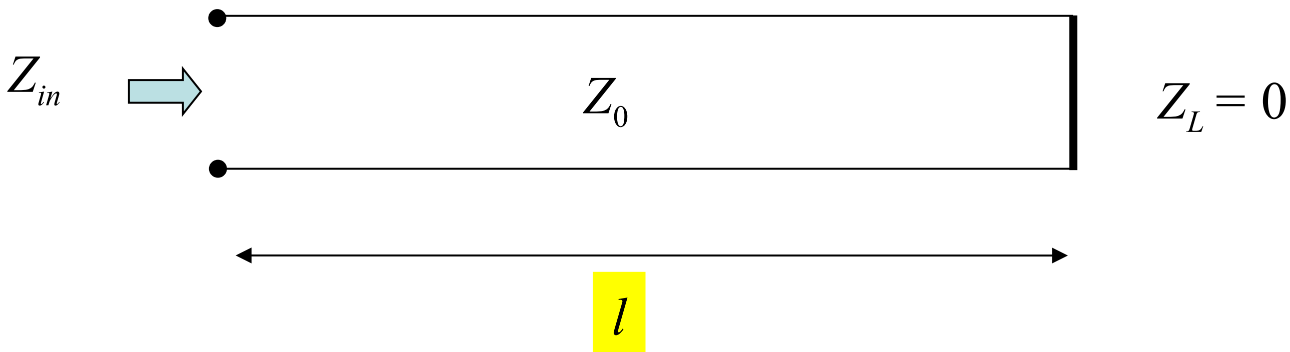
$$\beta = \omega\sqrt{LC} = \omega\sqrt{\mu\varepsilon} = \frac{2\pi}{\lambda}$$

$$\lambda = \lambda_0 / \sqrt{\varepsilon_r}$$

# Impedance (cont.)

## Special Cases of Lossless Line

### Short-circuit line




$$Z_{in} = Z_0 \left( \frac{\cancel{Z_L} + jZ_0 \tan(\beta l)}{Z_0 + j\cancel{Z_L} \tan(\beta l)} \right)$$

$$\boxed{Z_{in} = jZ_0 \tan(\beta l)}$$

or

$$Z_{in} = jZ_0 \tan(2\pi l / \lambda)$$

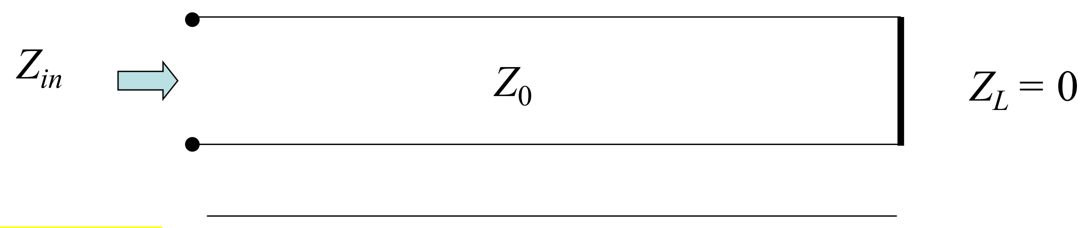
# Impedance (cont.)

## Short-circuit line

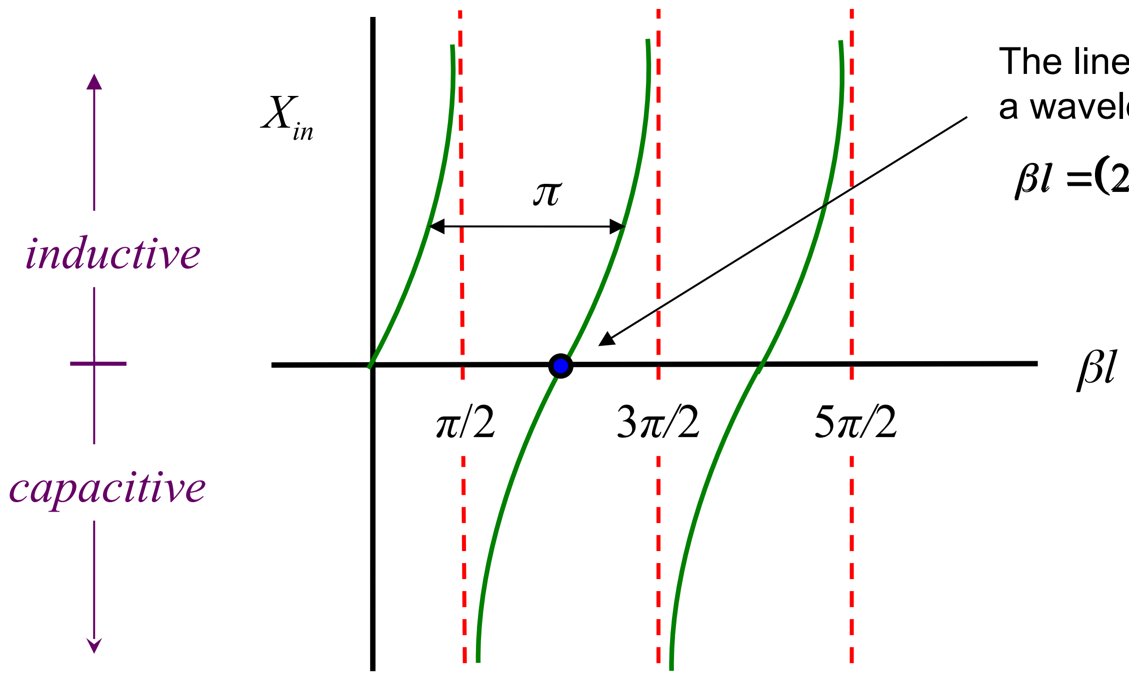
$$X_{in} = Z_0 \tan(\beta l)$$

or

$$X_{in} = Z_0 \tan(2\pi l / \lambda)$$




$Z_{in} = jX_{in}$



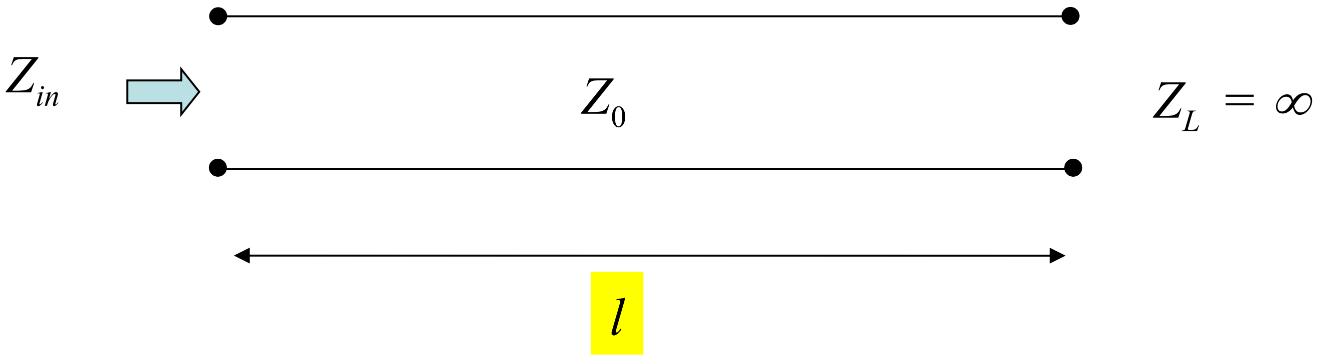

The line is one-half of a wavelength long.

$$\beta l = (2\pi / \lambda) l = \pi$$

# Impedance (cont.)

## Special cases of Lossless Line (cont.)

### Open-circuit line




$$Z_{in} = Z_0\left(\frac{Z_L + jZ_0\tan(\beta l)}{Z_0 + jZ_L\tan(\beta l)}\right) \Rightarrow Z_{in} = Z_0\left(\frac{1 + j\cancel{(Z_0/Z_L)}\tan(\beta l)}{\cancel{(Z_0/Z_L)} + j\tan(\beta l)}\right)$$

$$\boxed{Z_{in} = -jZ_0\cot(\beta l)}$$

or

$$Z_{in} = -jZ_0\cot(2\pi l/\lambda)$$

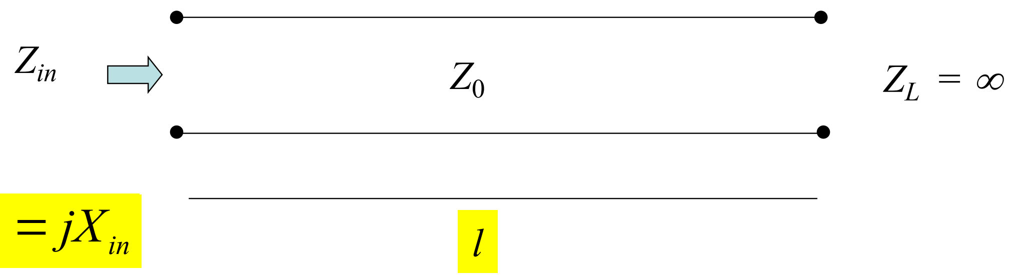
# Impedance (cont.)

Open-circuit line

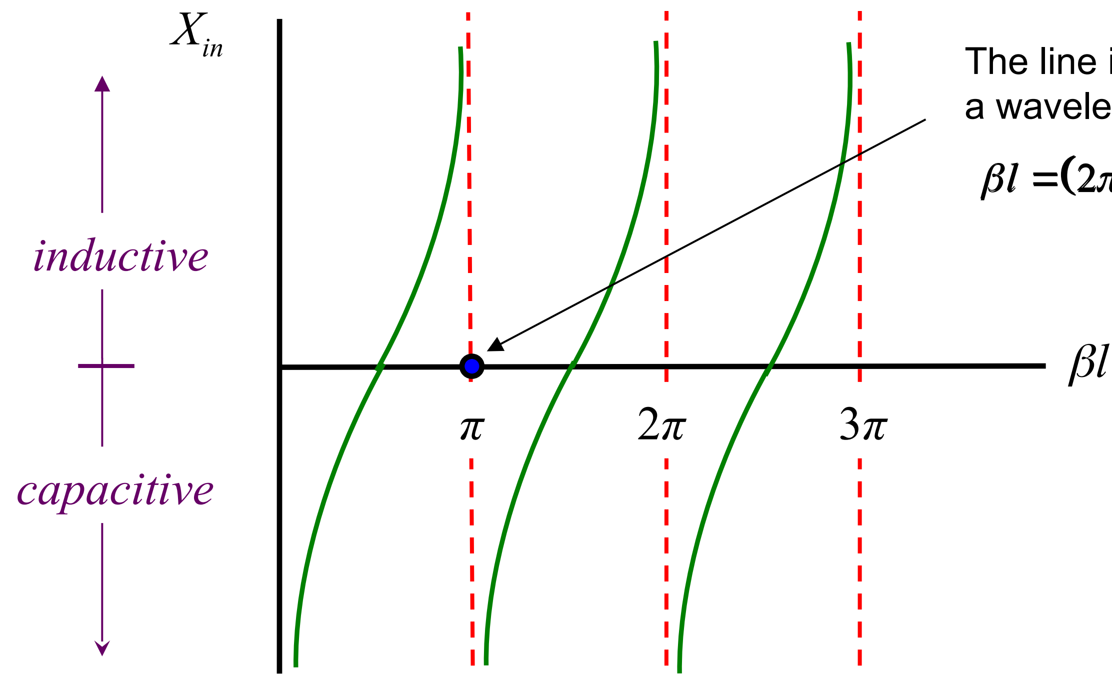
$$X_{in} = -Z_0 \cot(\beta l)$$

or

$$X_{in} = -Z_0 \cot(2\pi l / \lambda)$$




$$Z_{in} = jX_{in}$$




The line is one-half of a wavelength long.

$$\beta l = (2\pi / \lambda) l = \pi$$

# Filter Application

Microstrip filter (application of open-circuited “stubs”)